

Chapter 6: Advanced Optimization, in which we cover an advanced method for finding optimal solutions: simulated annealing. We also introduce the traveling salesman problem, a standard problem in advanced computer science.

Chapter 7: Geometry, in which we go over how to generate Voronoi diagrams, which can be useful in a variety of geometric applications.

Chapter 8: Language, in which we discuss how to intelligently add spaces to a text that's missing them, and how to intelligently suggest the next words in phrases.

Chapter 9: Machine Learning, in which we discuss decision trees, a fundamental machine learning method.

Chapter 10: Artificial Intelligence, in which we jump to an ambitious project: implementing an algorithm that can play games against us—and maybe even win. We start with a simple game, dots and boxes, and discuss how we could improve performance.

Chapter 11: Forging Ahead, in which we talk about how to progress to more advanced work related to algorithms. We discuss how to build a chatbot, and how to win a million dollars by creating a sudoku algorithm.

Setting Up the Environment

We'll implement the algorithms described in this book by using the Python language. Python is free and open source, and it runs on every major platform. You can use the following steps to install Python on Windows, macOS, and Linux.

Install Python on Windows

To install Python on Windows, follow these steps:

1. Open the page dedicated to the latest version of Python for Windows (make sure you include the final slash):
<https://www.python.org/downloads/windows/>.